Chapter 3: Python Data Types

Type	Draft
Status	Not started
Project	My Python Journey

Python Data Types – Overview

Definition: A data type defines what kind of value a variable stores.

Python has the following built-in data types:

1. Numeric Types

Used to store numbers.

Data type	Meaning	Example
`int`	Whole numbers	`age = 22`
`float`	Decimal numbers	`price = 99.5`
`complex`	Real and imaginary numbers	`x = 5 + 3j`

2. Text Type

Data type	Meaning	Example
`str`	Stores text or characters	`name = "Python"`

3. Boolean Type

Data type	Meaning	Example
`bool`	Stores True or False	`is_valid = True`

4. Sequence Types

Used to store multiple values.

Data type	Meaning	Example
`list`	Ordered, mutable collection	`[10, 20, 30]`
`tuple`	Ordered, immutable collection	`(10, 20, 30)`
`range`	Sequence of numbers	`range(1, 5)`

5. Mapping Type

Data type	Meaning	Example
`dict`	Stores key-value pairs	`{"name": "Ram"}`

6. Set Types

Used to store unique elements.

Data type	Meaning	Example
`set`	Mutable collection of unique elements	`{10, 20, 30}`
`frozenset`	Immutable collection of unique elements	`frozenset([10, 20])`

7. Binary Types

Used to store binary data.

Data type	Meaning
`bytes`	Immutable binary data
`bytearray`	Mutable binary data
`memoryview`	Provides access to binary data without copying

8. None Type

Data type	Meaning	Example
`NoneType`	Represents the absence of a value	`x = None`

Quick revision table

Remember these 8 categories

1. Numeric → `int`, `float`, `complex`
2. Text → `str`
3. Boolean → `bool`
4. Sequence → `list`, `tuple`, `range`
5. Mapping → `dict`
6. Set → `set`, `frozenset`
7. Binary → `bytes`, `bytearray`, `memoryview`
8. None → `NoneType`

Important: Use `type()` to check the data type of a variable.

```
x = 100print(type(x))
```

Output: `<class 'int'>`